

Distributional Semantic Models

Part 1: Introduction

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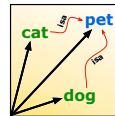
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<http://wordspace.collocations.de/doku.php/course:start>

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Outline

Introduction

The distributional hypothesis
Distributional semantic models
DSMs and semantic similarity
Three famous examples

Getting practical

Software and further information
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Meaning & distribution

- ▶ “Die Bedeutung eines Wortes liegt in seinem Gebrauch.”
— Ludwig Wittgenstein
☞ meaning = use = distribution in language
- ▶ “You shall know a word by the company it keeps!”
— J. R. Firth (1957)
☞ distribution = collocations = habitual word combinations
- ▶ Distributional hypothesis: difference of meaning correlates with difference of distribution (Zellig Harris 1954)
☞ semantic distance
- ▶ “What people know when they say that they know a word is not how to recite its dictionary definition – they know how to use it [...] in everyday discourse.” (Miller 1986)

What is the meaning of the mystery word?

Can we infer meaning from usage?

- ▶ He handed her her glass of ████████.
- ▶ Beef dishes are made to complement the ████████s.
- ▶ Nigel staggered to his feet, face flushed from too much ████████.
- ▶ Malbec, one of the lesser-known ████████ grapes, responds well to Australia's sunshine.
- ▶ I dined off bread and cheese and this excellent ████████.
- ▶ The drinks were delicious: blood-red ████████ as well as light, sweet Rhenish.

 ████████ is a heavy red alcoholic beverage made from grapes

All examples from British National Corpus (handpicked and slightly edited).

Word sketch of “cat”

Can we infer meaning from collocations (as Firth suggests)?

cat British National Corpus freq = 5381

<https://sketchengine.eu/>

object of 964 2.0	and/or 1056 1.7	pp_obj like-p 106 28.9	possessor 91 1.9	possession 232 4.7
skin 9 7.91	dog 208 8.49	grin 11 7.63	Schrödinger 8 10.87	cradle 24 9.91
diddle 7 7.85	cat 68 8.01	fight 9 4.62	witch 4 6.82	whisker 9 8.92
stroke 10 7.09	kitten 13 8.01	smile 4 4.24	gardener 4 6.0	paw 5 7.44
torture 5 6.57	fiddle 9 7.71	look 11 2.04	Henry 8 4.91	fur 9 7.14
feed 22 6.34	mouse 29 7.68		neighbour 5 4.28	tray 4 5.34
rain 4 6.3	monkey 15 7.55	pp among-p 17 14.8		tail 5 4.91
chase 9 6.27	budgie 4 6.74	pigeon 15 8.66		tongue 5 4.89
rescue 7 6.15	rabbit 12 6.48			ear 5 4.0

subject of 842 3.3	adj. subject of 142 2.6	pp_obj of-p 324 1.3	modifier 1622 1.2	modifies 610 0.5
purr 7 7.76	asleep 4 6.09	moral 4 7.06	pussy 76 10.42	flap 16 8.39
miaow 5 7.57	alive 4 5.06	breed 6 5.77	Cheshire 45 8.9	litter 15 8.15
mew 4 7.18	concerned 4 2.94	signal 4 3.89	stray 25 8.7	phobia 5 7.64
jump 20 6.95	black 4 2.36	sight 4 3.77	siamese 17 8.35	burglar 8 7.55
scratch 8 6.84	likely 4 1.96	species 5 3.36	tabby 17 8.35	faeces 6 7.47
leap 10 6.78		game 9 3.14	wild 53 7.94	assay 10 7.38
stalk 4 6.56		picture 6 2.99	pet 31 7.92	Hastings 7 6.91
react 4 5.33		death 7 2.71	tom 12 7.8	scan 4 6.59

A thought experiment: deciphering hieroglyphs

						
(knife)		51	20	84	0	3
(cat)		52	58	4	4	6
???		115	83	10	42	33
(boat)		59	39	23	4	0
(cup)		98	14	6	2	1
(pig)		12	17	3	2	9
(banana)		11	2	2	0	18

DSM Tutorial – Part 1

- Introduction
- The distributional hypothesis
- A thought experiment: deciphering hieroglyphs

2026-09-30

A thought experiment: deciphering hieroglyphs

	knife	cat	boat	cup	pig	banana
(knife)	51	20	84	0	3	0
(cat)	52	58	4	4	6	26
???	115	83	10	42	33	17
(boat)	59	39	23	4	0	0
(cup)	98	14	6	2	1	0
(pig)	12	17	3	2	9	27
(banana)	11	2	2	0	18	0

- Similarity scores are cosine similarities on sparse log-scaled frequencies ($\log(f + 1)$).

A thought experiment: deciphering hieroglyphs

(knife)	51	20	84	0	3	0
(cat)	52	58	4	4	6	26
???	115	83	10	42	33	17
(boat)	59	39	23	4	0	0
(cup)	98	14	6	2	1	0
(pig)	12	17	3	2	9	27
(banana)	11	2	2	0	18	0

$$\text{sim}(\text{hieroglyph of a dog}, \text{hieroglyph of a knife}) = 0.770$$

A thought experiment: deciphering hieroglyphs

(knife)	51	20	84	0	3	0
(cat)	52	58	4	4	6	26
???	115	83	10	42	33	17
(boat)	59	39	23	4	0	0
(cup)	98	14	6	2	1	0
(pig)	12	17	3	2	9	27
(banana)	11	2	2	0	18	0

$$\text{sim}(\text{hieroglyph of a dog}, \text{hieroglyph of a pig}) = 0.939$$

A thought experiment: deciphering hieroglyphs

(knife)	51	20	84	0	3	0
(cat)	52	58	4	4	6	26
???	115	83	10	42	33	17
(boat)	59	39	23	4	0	0
(cup)	98	14	6	2	1	0
(pig)	12	17	3	2	9	27
(banana)	11	2	2	0	18	0

$$\text{sim}(\text{hieroglyph of a dog}, \text{hieroglyph of a cat}) = 0.961$$

English as seen by the computer ...

	get 	see 	use 	hear 	eat 	kill
knife	51	20	84	0	3	0
cat	52	58	4	4	6	26
dog	115	83	10	42	33	17
boat	59	39	23	4	0	0
cup	98	14	6	2	1	0
pig	12	17	3	2	9	27
banana	11	2	2	0	18	0

verb-object counts from British National Corpus

Geometric interpretation

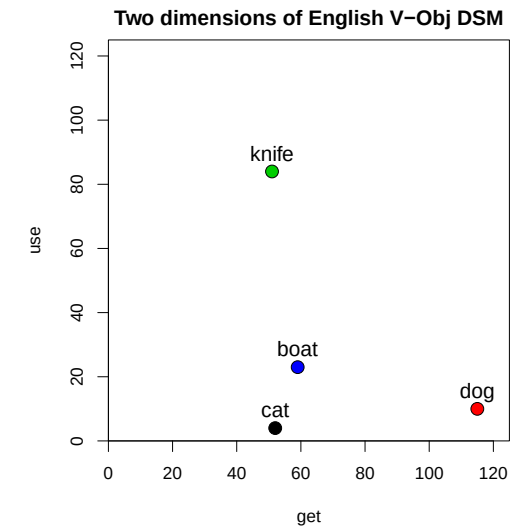
- ▶ row vector $\mathbf{x}_{\text{dog}}$ describes usage of word *dog* in the corpus
- ▶ can be seen as coordinates of point in n -dimensional Euclidean space

	get	see	use	hear	eat	kill
knife	51	20	84	0	3	0
cat	52	58	4	4	6	26
dog	115	83	10	42	33	17
boat	59	39	23	4	0	0
cup	98	14	6	2	1	0
pig	12	17	3	2	9	27
banana	11	2	2	0	18	0

co-occurrence matrix $\mathbf{M}$

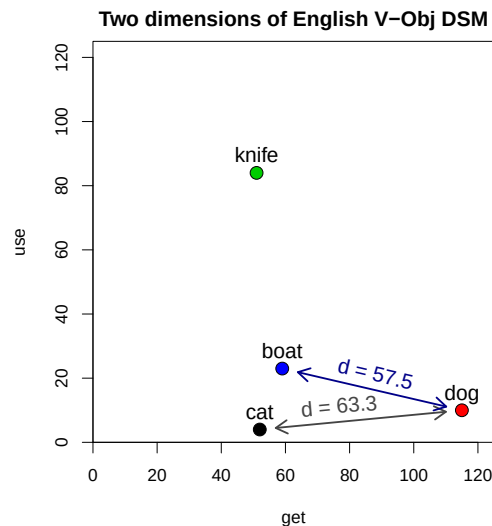
Geometric interpretation

- ▶ row vector $\mathbf{x}_{\text{dog}}$ describes usage of word *dog* in the corpus
- ▶ can be seen as coordinates of point in n -dimensional Euclidean space
- ▶ illustrated for two dimensions: *get* and *use*
- ▶ $\mathbf{x}_{\text{dog}} = (115, 10)$



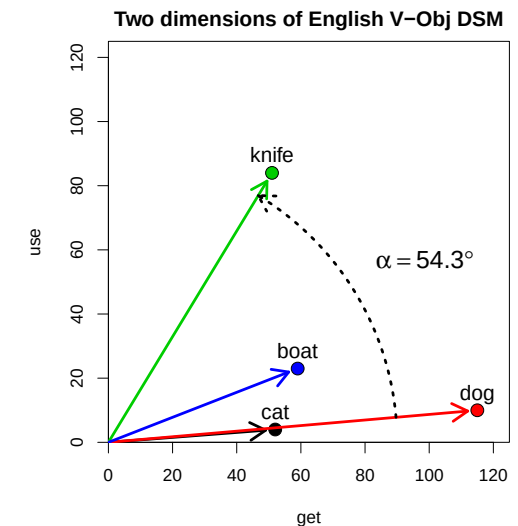
Geometric interpretation

- ▶ similarity = spatial proximity (Euclidean dist.)
- ▶ location depends on frequency of noun ($f_{\text{dog}} \approx 2.7 \cdot f_{\text{cat}}$)



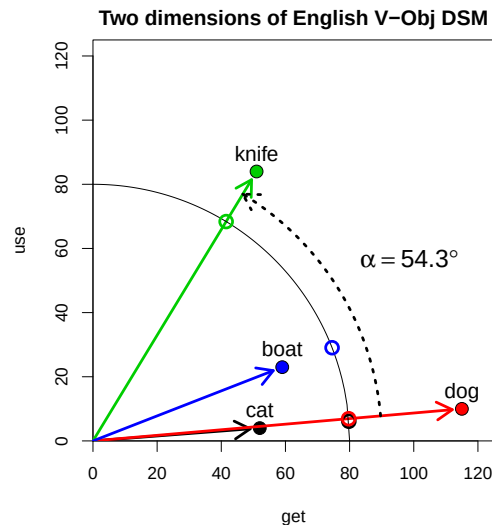
Geometric interpretation

- ▶ vector can also be understood as arrow from origin
- ▶ direction more important than location
- ▶ use angle α as distance measure



Geometric interpretation

- ▶ vector can also be understood as arrow from origin
- ▶ direction more important than location
- ▶ use angle α as distance measure
- ▶ or normalise length $\|x_{\text{dog}}\|$ of arrow



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General definition of DSMs

A **distributional semantic model** (DSM) is a scaled and/or transformed co-occurrence matrix **M**, such that each row **x** represents the distribution of a target term across contexts.

	get	see	use	hear	eat	kill
knife	0.027	-0.024	0.206	-0.022	-0.044	-0.042
cat	0.031	0.143	-0.243	-0.015	-0.009	0.131
dog	-0.026	0.021	-0.212	0.064	0.013	0.014
boat	-0.022	0.009	-0.044	-0.040	-0.074	-0.042
cup	-0.014	-0.173	-0.249	-0.099	-0.119	-0.042
pig	-0.069	0.094	-0.158	0.000	0.094	0.265
banana	0.047	-0.139	-0.104	-0.022	0.267	-0.042

Term = word, lemma, phrase, morpheme, word pair, ...

Nearest neighbours

DSM based on verb-object relations from BNC, reduced to 100 dim. with SVD

Neighbours of **trousers** (cosine angle):

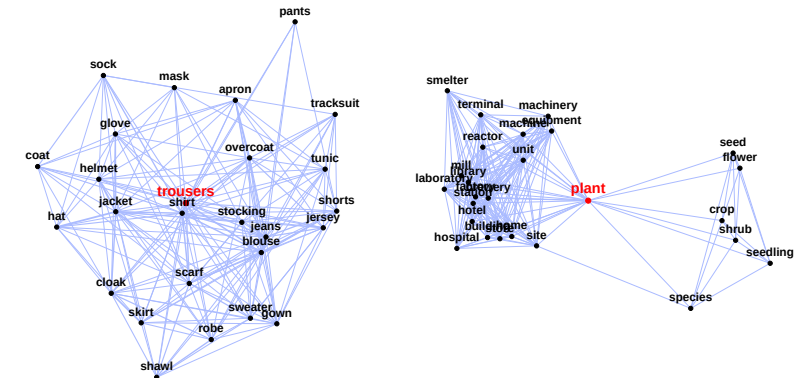
shirt (18.5), blouse (21.9), scarf (23.4), jeans (24.7), skirt (25.9), sock (26.2), shorts (26.3), jacket (27.8), glove (28.1), coat (28.8), cloak (28.9), hat (29.1), tunic (29.3), overcoat (29.4), pants (29.8), helmet (30.4), apron (30.5), robe (30.6), mask (30.8), tracksuit (31.0), jersey (31.6), shawl (31.6), ...

Neighbours of **rage** (cosine angle):

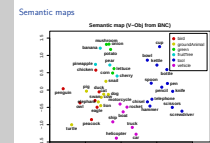
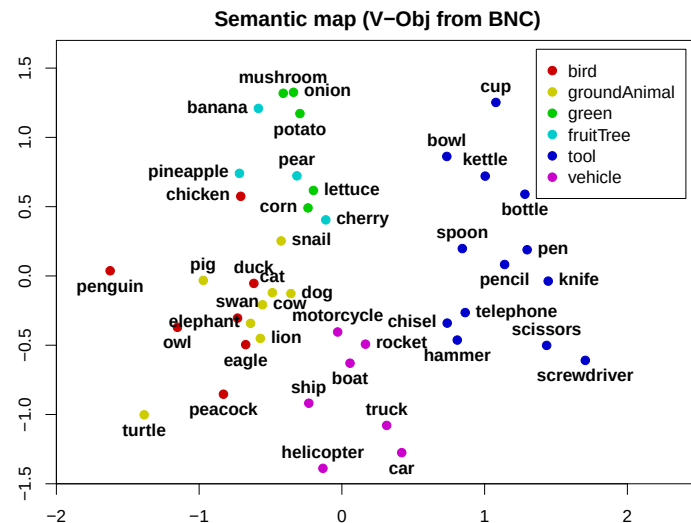
anger (28.5), fury (32.5), sadness (37.0), disgust (37.4), emotion (39.0), jealousy (40.0), grief (40.4), irritation (40.7), revulsion (40.7), scorn (40.7), panic (40.8), bitterness (41.6), resentment (41.8), indignation (41.9), excitement (42.0), hatred (42.5), envy (42.8), disappointment (42.9), ...

1. Neighbours and neighbourhood plots from BNC verb-object DSM, reduced to 100 dimensions by SVD.

Nearest neighbours with similarity graph

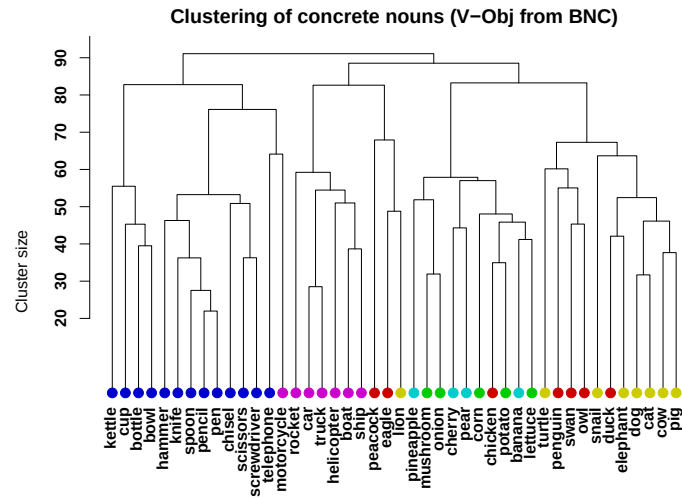


Semantic maps

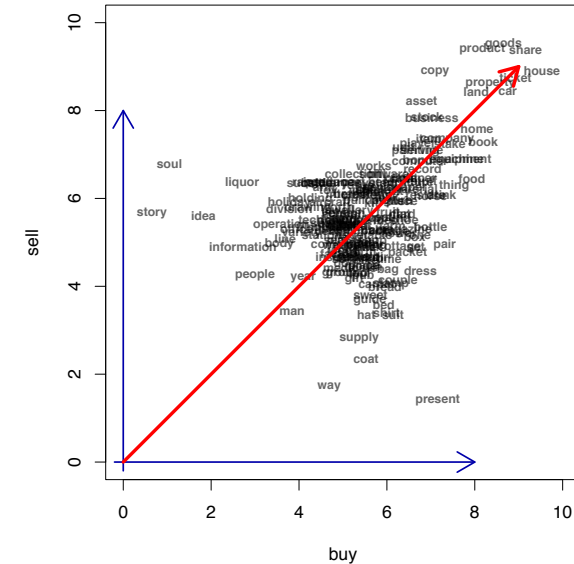


1. Roughly horizontal axis separates natural objects (left) from artifacts (right), or animate vs. inanimate. There is a clear boundary between the two groups.
2. Orthogonal axis separates moving things (bottom) from motionless ones (top).

Clustering



Latent “meaning” dimensions



DSM vectors as word embeddings

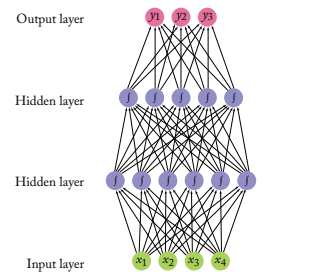
DSM vector as sub-symbolic meaning representation

- ▶ feature vector for machine learning algorithm
- ▶ input for neural network
- ▶ such **distributed** representations are known as **embeddings**

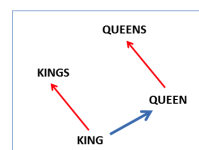
🖱 embeddings $\neq$ distributional

Computation in semantic space

- ▶ find meaningful subdimensions in DSM space ($\rightarrow$ correlation)
- ▶ linear operations on vectors



(Goldberg 2017, Fig. 4.2)

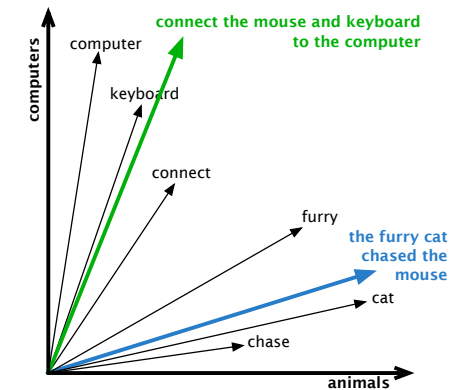


(Mikolov et al. 2013c, Fig. 2)

DSM vectors as context representation

Context vectors for word tokens (Schütze 1998)

- ▶ **bag-of-words** approach: centroid of all context words in the sentence
- ▶ application to WSD



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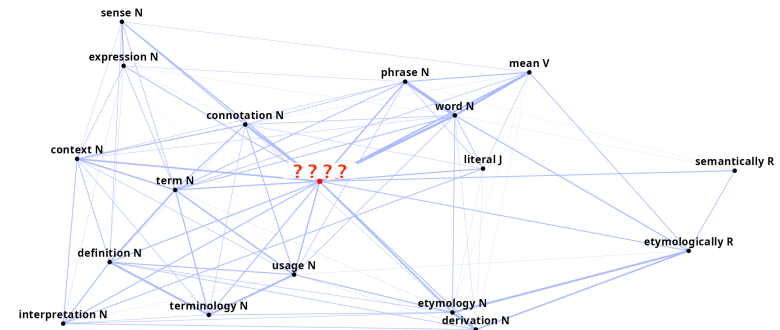
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Distributional similarity as semantic similarity

- ▶ DSM similarity as a **quantitative notion** (Harris 1954)
 - ▶ if **a** is closer to **b** than to **c** in the distributional vector space, then *a* is more semantically similar to *b* than to *c*
- ▶ DSM similarity as a **graded notion**, differently from **categorical** nature of most theoretical accounts
- ▶ DSM similarity as the empirical correlate of a **heterogeneous set of phenomena**
 - ... which we may want to tease apart!
- ▶ DSM similarity is **symmetric** – cognition is not
 - ... can we fix this?

Inverse distributional semantics

Which word occurs in similar contexts as the ones displayed in this graph?



... look at the neighbours:
 is there one notion of similarity that explains all of them?

Characterising DSM similarity

- ▶ DSMs are often thought to represent **taxonomic** similarity
 - ▶ words that tend to occur in the same contexts
- ▶ Words that share many contexts share many properties (attributes) and are thus **taxonomically/ontologically similar**
 - ▶ synonyms (*rhino/rhinoceros*)
 - ▶ antonyms and values on a scale (*good/bad*)
 - ▶ co-hyponyms (*rock/jazz*)
 - ▶ hyper- and hyponyms (*rock/basalt*)
- ▶ Taxonomic similarity is seen as the **fundamental semantic relation** organising the vocabulary of a language, allowing categorisation, generalisation and inheritance...

Is distributional similarity *just* taxonomic?

Nearest DSM neighbors have different types of **semantic relations**.

car (BNC, L2/R2 span)

- ▶ van **co-hyponym**
- ▶ vehicle **hyperonym**
- ▶ truck **co-hyponym**
- ▶ motorcycle **co-hyponym**
- ▶ driver **related entity**
- ▶ motor **part**
- ▶ lorry **co-hyponym**
- ▶ motorist **related entity**
- ▶ cavalier **hyponym**
- ▶ bike **co-hyponym**

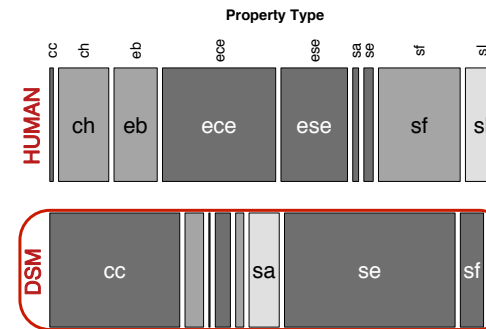
car (BNC, L30/R30 span)

- ▶ drive **function**
- ▶ park **typical action**
- ▶ bonnet **part**
- ▶ windscreen **part**
- ▶ hatchback **part**
- ▶ headlight **part**
- ▶ jaguar **hyponym**
- ▶ garage **location**
- ▶ cavalier **hyponym**
- ▶ tyre **part**

<http://clic.cimec.unitn.it/infomap-query/>

Is distributional similarity *just* taxonomic?

Manual annotation: what are the properties of *car*? Humans vs DSM



Taxonomic category:

cc (co-)hyponym *truck*
ch hypernym *vehicle*

Properties of entity:

eb typical behaviour
ece ext. component *wheel*
ese surf. property *smooth*

Situationally associated:

sa action *park*
se other entity *traffic light*
sf function *drive*
sl location *garage*
sp participant *driver*

Task: humans: given a word, generate properties; DSM: generate top 10 neighbors. Items: 44 concrete English nouns (Baroni & Lenci 2008).

DSM similarity: concepts & terminology

Attributional similarity vs. semantic relatedness

- ▶ **Attributional similarity** ($\leftarrow$ taxonomic) – two words sharing a large number of salient features (attributes)
 - ▶ synonymy (*car/automobile*)
 - ▶ co-hyponymy (*car/van/truck*)
 - ▶ hyperonymy (*car/vehicle*)
 - ☞ subset $\rightarrow$ distributional inclusion (Lenci & Benotto 2012)
 - ▶ antonymy (*hot/cold*)
 - ☞ intuitively they are the opposite of similar, and yet ...
- ▶ **Semantic relatedness** (Budanitsky & Hirst 2006) – two words semantically associated without necessarily being similar
 - ▶ function (*car/drive*)
 - ▶ meronymy (*car/tyre*)
 - ▶ location (*car/road*)
 - ▶ attribute (*car/fast*)

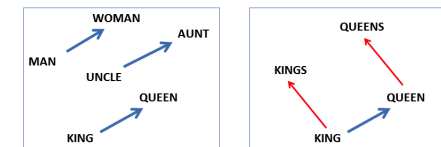
☞ Why similar in DSMs? co-occur $\rightarrow$ overlapping context spans

DSM similarity: concepts & terminology

Attributional vs. relational similarity

- ▶ **Attributional similarity** ($\leftarrow$ taxonomic) – two words sharing a large number of salient features (attributes)
 - ▶ synonymy (*car/automobile*)
 - ▶ co-hyponymy (*car/van/truck*)
 - ▶ hyperonymy (*car/vehicle*)
- ▶ **Relational similarity** (Turney 2006) – similar relation between pairs of words (analogy)
 - ▶ policeman : gun :: teacher : book
 - ▶ mason : stone :: carpenter : wood
 - ▶ traffic : street :: water : riverbed

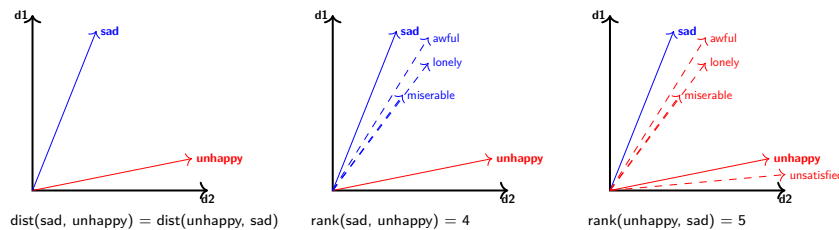
☞ Textbook example of neural embeddings application



Problem: symmetry in DSM similarity

Symmetric similarity does not fit all phenomena

- Motivation: cognitive processes are notoriously asymmetric (and so is human intuition about semantic similarity)
- Solution: use asymmetric measure of distributional similarity
- Solution: use neighbour rank instead of similarity score
 - makes predictions comparable across models and measures
 - intuition: different density of semantic neighbourhood



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Latent Semantic Analysis (Landauer & Dumais 1997)

- Corpus: 30,473 articles from Grolier's *Academic American Encyclopedia* (4.6 million words in total)
 - articles were limited to first 2,000 characters
- Word-article frequency matrix for 60,768 words
 - row vector shows frequency of word in each article
- Logarithmic frequencies scaled by word entropy
- Reduced to 300 dim. by singular value decomposition (SVD)
 - borrowed from LSI (Dumais *et al.* 1988)
 - central claim: SVD reveals latent semantic features, not just a data reduction technique
- Evaluated on TOEFL synonym test (80 items)
 - LSA model achieved 64.4% correct answers
 - also simulation of learning rate based on TOEFL results

Word Space (Schütze 1992, 1993, 1998)

- Corpus: ≈ 60 million words of news messages
 - from the *New York Times* News Service
- Word-word co-occurrence matrix
 - 20,000 target words & 2,000 context words as features
 - row vector records how often each context word occurs close to the target word (co-occurrence)
 - co-occurrence window: left/right 50 words (Schütze 1998) or ≈ 1000 characters (Schütze 1992)
- Rows weighted by inverse document frequency (tf.idf)
- Context vector = centroid of word vectors (bag-of-words)
 - goal: determine “meaning” of a context
- Reduced to 100 SVD dimensions (mainly for efficiency)
- Evaluated on unsupervised word sense induction by clustering of context vectors (for an ambiguous word)
 - induced word senses improve information retrieval performance

HAL (Lund & Burgess 1996)

- ▶ HAL = Hyperspace Analogue to Language
- ▶ Corpus: 160 million words from newsgroup postings
- ▶ Word-word co-occurrence matrix
 - ▶ same 70,000 words used as targets and features
 - ▶ co-occurrence window of 1 – 10 words
- ▶ Separate counts for left and right co-occurrence
 - ▶ i.e. the context is *structured*
- ▶ In later work, co-occurrences are weighted by (inverse) distance (Li *et al.* 2000)
 - ▶ but no dimensionality reduction
- ▶ Applications include construction of semantic vocabulary maps by multidimensional scaling to 2 dimensions

HAL (Lund & Burgess 1996)

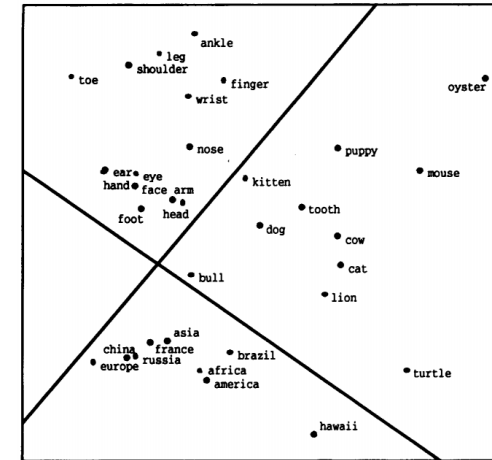


Figure 2. Multidimensional scaling of co-occurrence vectors.

word2vec embeddings (Mikolov *et al.* 2013a,b)

- ▶ Embeddings are learned by a (simple) neural model rather than obtained directly as distributional vectors
 - ▶ known as **predict** rather than **count** models (Baroni *et al.* 2014)
- ▶ Corpus: Google News, 6 billion words
- ▶ Learning objective of the skip-gram model (SGNS)
 - ▶ predict conditional probability distribution $\Pr(c | t)$ for collocational profile of target word t
 - ▶ based on learned target embeddings $\mathbf{w}_t$ and (ephemeral) collocate embeddings $\mathbf{v}_c$
 - ▶ can be understood as factorisation of co-occurrence matrix (Levy & Goldberg 2014; Pennington *et al.* 2014)
- ▶ Evaluation focuses on syntactic and semantic analogy tasks

Many parameters ...

- ▶ Enormous range of DSM parameters and applications
- ▶ Examples showed four entirely different models, each tuned to its particular application
- ➡ Need overview of DSM parameters & understand their effects
 - ▶ part 2: The parameters of a DSM
 - ▶ part 3: Evaluating DSM representations
 - ▶ part 4: Understanding distributional semantics
 - ▶ part 5: Mathematical foundations (matrix algebra)
- ➡ Distributional semantics is an empirical science

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Software packages

Infomap NLP	C	<i>classical LSA-style DSM</i>
HiDEx	C++	<i>re-implementation of the HAL model (Lund & Burgess 1996)</i>
SemanticVectors	Java	<i>scalable architecture based on random indexing representation</i>
S-Space	Java	<i>complex object-oriented framework</i>
JoBimText	Java	<i>UIMA / Hadoop framework</i>
Gensim	Python	<i>complex framework, focus on parallelization and out-of-core algorithms</i>
Vecto	Python	<i>framework for count & predict models</i>
DISSECT	Python	<i>user-friendly, designed for research on compositional semantics</i>
wordspace	R	<i>interactive research laboratory, but scales to real-life data sets</i>
text2vec	R	<i>GloVe embeddings & topic models</i>

click on package name to open Web page

Some applications in computational linguistics

- Query expansion in information retrieval (Grefenstette 1994)
- Unsupervised part-of-speech induction (Schütze 1995)
- Word sense disambiguation (Schütze 1998; Rapp 2004b)
- Thesaurus compilation (Lin 1998; Rapp 2004a)
- Attachment disambiguation (Pantel & Lin 2000)
- Probabilistic language models (Bengio *et al.* 2003)
- Translation equivalents (Sahlgren & Karlgren 2005)
- Ontology & wordnet expansion (Pantel *et al.* 2009)
- Language change (Sagi *et al.* 2009; Hamilton *et al.* 2016)
- Multiword expressions (Kiehl & Clark 2013)
- Analogies (Turney 2013; Gladkova *et al.* 2016)
- Sentiment analysis (Rothe & Schütze 2016; Yu *et al.* 2017)
- Input representation for neural networks & machine learning

Further information

- Handouts & other materials available from wordspace wiki at <http://wordspace.collocations.de/>
- Tutorial is open source (CC) and can be downloaded from <http://r-forge.r-project.org/projects/wordspace/>

Further information

- ▶ Review papers on distributional semantics
 - ▶ Turney, Peter D. and Pantel, Patrick (2010). *From frequency to meaning: Vector space models of semantics*. *Journal of Artificial Intelligence Research*, 37, 141–188.
 - ▶ Erk, Katrin (2012). *Vector Space Models of Word Meaning and Phrase Meaning: A Survey*. *Language and Linguistics Compass*, 6-1, 635–653.
 - ▶ Boleda, Gemma (2020). *Distributional Semantics and Linguistic Theory*. *Annual Review of Linguistics*, 6-1, 213–234.
- ▶ Textbook on distributional semantics
 - ▶ Lenci, Alessandro and Sahlgren, Magnus (2023). *Distributional Semantics*. Studies in Natural Language Processing. Cambridge University Press, Cambridge.

Prepare to get your hands dirty ...

- ▶ We will use the statistical programming environment **R** as a toy laboratory in this tutorial
 - ▶ but one that scales to real-life applications

Software installation

- ▶ **R** version 4.0 or newer from <http://www.r-project.org/>
- ▶ **RStudio** from <http://www.rstudio.com/>
- ▶ R packages from CRAN (through RStudio menu)
 - ▶ `sparsesvd`, `wordspace`
 - ▶ recommended: `e1071`, `rsparse`, `Rtsne`, `uwot`
 - ▶ optional: `tm`, `quanteda`, `data.table`, `wordcloud`, `shiny`, `spacyr`, `udpipe`, `coreNLP`
- ▶ Get data sets, precompiled DSMs and `wordspaceEval` package (with some non-public data sets) from <http://wordspace.collocations.de/doku.php/course:material>

Outline

Introduction

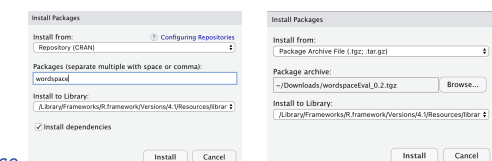
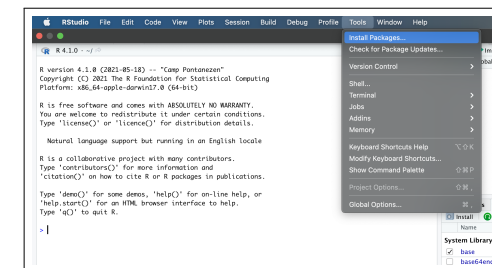
The distributional hypothesis
 Distributional semantic models
 DSMs and semantic similarity
 Three famous examples

Getting practical

Software and further information
 R as a (toy) laboratory

Prepare to get your hands dirty ...

Installing `wordspace` and `wordspaceEval` in RStudio



`wordspace`

`wordspaceEval`

Prepare to get your hands dirty ...

Setting up a working directory and RStudio project

- ▶ Create a separate **directory** (folder) for this course
 - ▶ subdirectory **models** for pre-compiled DSMs (large files)
 - ▶ subdirectory **data** for other data files
- ▶ Recommended: set up **RStudio project** for the course
 - ▶ click *New Project* (top right corner), then *Existing Directory*
 - ▶ choose the course directory you've just created
 - ▶ this will be set as your R working directory within the project!
 - ▶ you can easily switch between different RStudio projects
- ▶ Alternatively: set working directory at start of session
 - ▶ e.g. `setwd("~/zellig/Desktop/dsm_tutorial")`
- ▶ Work with **R scripts** rather than in interactive console
 - ▶ RStudio: add *R Script* from drop-down menu in top left corner
 - ▶ we provide example scripts for each part of the tutorial

First steps in R

Start each session by loading the wordspace package.

```
> library(wordspace)
```

The package includes various example data sets, some of which should look familiar to you.

```
> DSM_HieroglyphsMatrix
      get see use hear eat kill
knife  51  20  84   0   3   0
cat     52  58   4   4   6  26
dog    115  83  10  42  33  17
boat    59  39  23   4   0   0
cup     98  14   6   2   1   0
pig     12  17   3   2   9  27
banana  11   2   2   0  18   0
```

Term-term matrix

Term-term matrix records co-occurrence frequencies with feature terms for each target term

```
> DSM_TermTermMatrix
```

	breed	tail	feed	kill	important	explain	likely
cat	83	17	7	37	–	1	–
dog	561	13	30	60	1	2	4
animal	42	10	109	134	13	5	5
time	19	9	29	117	81	34	109
reason	1	–	2	14	68	140	47
cause	–	1	–	4	55	34	55
effect	–	–	1	6	60	35	17

Term-context matrix

Term-context matrix records frequency of term in each individual context (e.g. sentence, document, Web page, encyclopaedia article)

```
> DSM_TermContextMatrix
```

	Felidae	Pet	Feral	Bloat	Philosophy	Kant	Back pain
cat	10	10	7	–	–	–	–
dog	–	10	4	11	–	–	–
animal	2	15	10	2	–	–	–
time	1	–	–	–	2	1	–
reason	–	1	–	–	1	4	1
cause	–	–	–	2	1	2	6
effect	–	–	–	1	–	1	–

Some basic operations on a DSM matrix

```
# apply log-transformation to de-skew co-occurrence frequencies
> M <- log2(DSM_HieroglyphsMatrix + 1) # see part 2
> round(M, 3)

# compute distributional similarity (cosine measure)
> pair.distances("dog", "cat", M, convert=FALSE)
dog/cat
0.9610952

# find nearest neighbours (defaults to angles)
> nearest.neighbours(M, "dog", n=3)
      cat      pig      cup
16.03458 20.08826 31.77784

> plot(nearest.neighbours(M, "dog", n=3, dist.matrix=TRUE))
```

Playing with a larger model

Term-term matrix, dimensionality-reduced, built from Web texts for target words in the format *lemma_POS* (e.g. banana_N)

```
> DSM_Vectors
> View(DSM_Vectors)
```

Let's inspect some nearest neighbors:

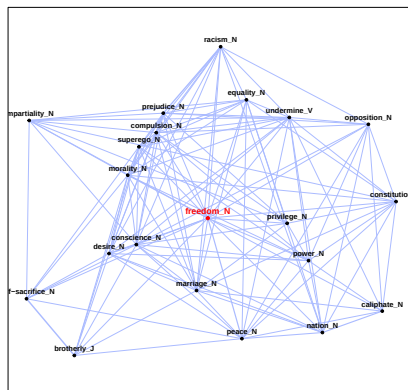
```
> nearest.neighbours(DSM_Vectors, "banana_N", n=4)
coconut_N  pineapple_N  watermelon_N  bean_N
10.86118    12.60826    13.35160    13.79671
```

```
> nearest.neighbours(DSM_Vectors, "freedom_N", n=4)
peace_N    morality_N    equality_N    conscience_N
30.13420    34.18397    34.23418    34.23894
```

Playing with a larger model

Or create a semantic map for a word we are interested in:

```
> plot(nearest.neighbours(DSM_Vectors, "freedom_N", n=20,
  dist.matrix=TRUE))
```



... and with an even larger model

You can download several **large pre-compiled DSMs** from the course wiki, which represent different parameters of the co-occurrence matrix (→ part 2).

- ▶ e.g. WP500_DepFilter_Lemma.rda
- ▶ download this file to subdirectory models

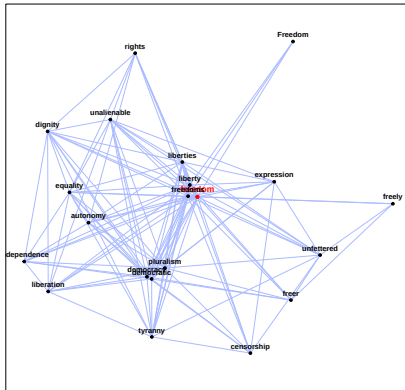
```
> load("models/WP500_DepFilter_Lemma.rda", verbose=TRUE)
Loading objects:
WP500_DepFilter_Lemma
> model <- WP500_DepFilter_Lemma # assign to a shorter name
```

Now try the semantic map again:

```
> plot(nearest.neighbours(model, "freedom_N", n=20,
  dist.matrix=TRUE))
```

Freedom in a neural embedding model: word2vec

```
> load("GoogleNews300_wf200k.rda", verbose=TRUE)
> embeddings <- GoogleNews300_wf200k.rda
> plot(nearest.neighbours(embeddings, "freedom", n=20,
                           dist.matrix=TRUE))
```



References I

- Baroni, Marco and Lenci, Alessandro (2008). Concepts and properties in word spaces. *Italian Journal of Linguistics*, 20(1).
- Baroni, Marco; Dinu, Georgiana; Kruszewski, Germán (2014). Don't count, predict! A systematic comparison of context-counting vs. context-predicting semantic vectors. In *Proceedings of the 52nd Annual Meeting of the Association for Computational Linguistics*, pages 238–247, Baltimore, MD.
- Bengio, Yoshua; Ducharme, Réjean; Vincent, Pascal; Jauvin, Christian (2003). A neural probabilistic language model. *Journal of Machine Learning Research*, 3, 1137–1155.
- Boleda, Gemma (2020). Distributional semantics and linguistic theory. *Annual Review of Linguistics*, 6(1), 213–234.
- Budanitsky, Alexander and Hirst, Graeme (2006). Evaluating WordNet-based measures of lexical semantic relatedness. *Computational Linguistics*, 32(1), 13–47.
- Dumais, S. T.; Furnas, G. W.; Landauer, T. K.; Deerwester, S.; Harshman, R. (1988). Using latent semantic analysis to improve access to textual information. In *CHI '88: Proceedings of the SIGCHI conference on Human factors in computing systems*, pages 281–285.
- Erk, Katrin (2012). Vector space models of word meaning and phrase meaning: A survey. *Language and Linguistics Compass*, 6(10), 635–653.

Explorations

While you wait for Part 2 of the tutorial,
you can explore some DSM similarity networks online:

- ▶ <https://corpora.linguistik.uni-erlangen.de/shiny/wordspace/>
- ▶ built in R with `wordspace` and `shiny`

References II

- Firth, J. R. (1957). A synopsis of linguistic theory 1930–55. In *Studies in linguistic analysis*, pages 1–32. The Philological Society, Oxford.
- Gladkova, Anna; Drozd, Aleksandr; Matsuoka, Satoshi (2016). Analogy-based detection of morphological and semantic relations with word embeddings: what works and what doesn't. In *Proceedings of the NAACL Student Research Workshop*, pages 8–15, San Diego, California.
- Goldberg, Yoav (2017). *Neural Network Methods for Natural Language Processing*. Number 37 in Synthesis Lectures on Human Language Technologies. Morgan & Claypool.
- Grefenstette, Gregory (1994). *Explorations in Automatic Thesaurus Discovery*, volume 278 of *Kluwer International Series in Engineering and Computer Science*. Springer, Berlin, New York.
- Hamilton, William L.; Leskovec, Jure; Jurafsky, Dan (2016). Diachronic word embeddings reveal statistical laws of semantic change. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1489–1501, Berlin, Germany.
- Harris, Zellig (1954). Distributional structure. *Word*, **10**(23), 146–162.

References III

- Kiela, Douwe and Clark, Stephen (2013). Detecting compositionality of multi-word expressions using nearest neighbours in vector space models. In *Proceedings of the 2013 Conference on Empirical Methods in Natural Language Processing (EMNLP 2013)*, pages 1427–1432, Seattle, WA.
- Landauer, Thomas K. and Dumais, Susan T. (1997). A solution to Plato's problem: The latent semantic analysis theory of acquisition, induction and representation of knowledge. *Psychological Review*, **104**(2), 211–240.
- Lenci, Alessandro and Benotto, Giulia (2012). Identifying hypernyms in distributional semantic spaces. In **SEM 2012: The First Joint Conference on Lexical and Computational Semantics – Volume 1: Proceedings of the main conference and the shared task, and Volume 2: Proceedings of the Sixth International Workshop on Semantic Evaluation (SemEval 2012)*, pages 75–79, Montréal, Canada. Association for Computational Linguistics.
- Lenci, Alessandro and Sahlgren, Magnus (2023). *Distributional Semantics*. Studies in Natural Language Processing. Cambridge University Press, Cambridge.
- Levy, Omer and Goldberg, Yoav (2014). Neural word embedding as implicit matrix factorization. In *Proceedings of Advances in Neural Information Processing Systems 27*, pages 2177–2185. Curran Associates, Inc.

References V

- Mikolov, Tomas; Yih, Wen-tau; Zweig, Geoffrey (2013c). Linguistic regularities in continuous space word representations. In *Proceedings of the 2013 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 746–751, Atlanta, GA.
- Miller, George A. (1986). Dictionaries in the mind. *Language and Cognitive Processes*, **1**, 171–185.
- Pantel, Patrick and Lin, Dekang (2000). An unsupervised approach to prepositional phrase attachment using contextually similar words. In *Proceedings of the 38th Annual Meeting of the Association for Computational Linguistics*, Hongkong, China.
- Pantel, Patrick; Crestan, Eric; Borkovsky, Arkady; Popescu, Ana-Maria; Vyas, Vishnu (2009). Web-scale distributional similarity and entity set expansion. In *Proceedings of the 2009 Conference on Empirical Methods in Natural Language Processing*, pages 938–947, Singapore.
- Pennington, Jeffrey; Socher, Richard; Manning, Christopher D. (2014). GloVe: Global vectors for word representation. In *Proceedings of EMNLP 2014*.
- Rapp, Reinhard (2004a). A freely available automatically generated thesaurus of related words. In *Proceedings of the 4th International Conference on Language Resources and Evaluation (LREC 2004)*, pages 395–398.

References IV

- Li, Ping; Burgess, Curt; Lund, Kevin (2000). The acquisition of word meaning through global lexical co-occurrences. In E. V. Clark (ed.), *The Proceedings of the Thirtieth Annual Child Language Research Forum*, pages 167–178. Stanford Linguistics Association.
- Lin, Dekang (1998). Automatic retrieval and clustering of similar words. In *Proceedings of the 17th International Conference on Computational Linguistics (COLING-ACL 1998)*, pages 768–774, Montreal, Canada.
- Lund, Kevin and Burgess, Curt (1996). Producing high-dimensional semantic spaces from lexical co-occurrence. *Behavior Research Methods, Instruments, & Computers*, **28**(2), 203–208.
- Mikolov, Tomas; Sutskever, Ilya; Chen, Kai; Corrado, Greg S.; Dean, Jeff (2013a). Distributed representations of words and phrases and their compositionality. In C. J. C. Burges, L. Bottou, M. Welling, Z. Ghahramani, and K. Q. Weinberger (eds.), *Proceedings of Advances in Neural Information Processing Systems 26 (NIPS 2013)*, pages 3111–3119. Curran Associates, Inc.
- Mikolov, Tomas; Chen, Kai; Corrado, Greg; Dean, Jeffrey (2013b). Efficient estimation of word representations in vector space. In *Workshop Proceedings of the International Conference on Learning Representations 2013*.

References VI

- Rapp, Reinhard (2004b). A practical solution to the problem of automatic word sense induction. In *Proceedings of the ACL-2004 Interactive Posters and Demonstrations Sessions*, pages 194–197, Barcelona, Spain. Association for Computational Linguistics.
- Rothe, Sascha and Schütze, Hinrich (2016). Word embedding calculus in meaningful ultradense subspaces. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 512–517, Berlin, Germany.
- Sagi, Eyal; Kaufmann, Stefan; Clark, Brady (2009). Semantic density analysis: Comparing word meaning across time and phonetic space. In *Proceedings of the Workshop on Geometrical Models of Natural Language Semantics (GEMS)*, pages 104–111, Athens, Greece.
- Sahlgren, Magnus and Karlgren, Jussi (2005). Automatic bilingual lexicon acquisition using random indexing of parallel corpora. *Natural Language Engineering*, **11**, 327–341.
- Schütze, Hinrich (1992). Dimensions of meaning. In *Proceedings of Supercomputing '92*, pages 787–796, Minneapolis, MN.
- Schütze, Hinrich (1993). Word space. In *Proceedings of Advances in Neural Information Processing Systems 5*, pages 895–902, San Mateo, CA.

References VII

- Schütze, Hinrich (1995). Distributional part-of-speech tagging. In *Proceedings of the 7th Conference of the European Chapter of the Association for Computational Linguistics (EACL 1995)*, pages 141–148.
- Schütze, Hinrich (1998). Automatic word sense discrimination. *Computational Linguistics*, **24**(1), 97–123.
- Turney, Peter D. (2006). Similarity of semantic relations. *Computational Linguistics*, **32**(3), 379–416.
- Turney, Peter D. (2013). Distributional semantics beyond words: Supervised learning of analogy and paraphrase. *Transactions of the Association for Computational Linguistics*, **1**, 353–366.
- Turney, Peter D. and Pantel, Patrick (2010). From frequency to meaning: Vector space models of semantics. *Journal of Artificial Intelligence Research*, **37**, 141–188.
- Yu, Liang-Chih; Wang, Jin; Lai, K. Robert; Zhang, Xuejie (2017). Refining word embeddings for sentiment analysis. In *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*, pages 534–539, Copenhagen, Denmark.